

Sethu Vardhan Valluri

Full Stack Developer | AI Application Developer | Software Engineer

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🐙 GitHub | 🔗 LinkedIn | 🏆 HackerRank | 🍳 CodeChef | 📖 LeetCode | 🇮🇳 Tanuku, India | 📁 Portfolio

SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in Full Stack Development, AI applications, Computer Networking, VLSI, and Embedded Systems. Skilled in React.js, Next.js, Angular, Node.js, Express.js, FastAPI, Java, Python, SQL/NoSQL databases, REST APIs, and AI/LLM integration. Strong foundation in TCP/IP, OSI, IPv4/IPv6, subnetting, VLANs, routing, switching, and network security. Academic experience in Verilog HDL, RTL design, AMBA protocols, AHB-to-APB bridge design, and Synopsys Sentaurus TCAD. Strong problem-solving, analytical, teamwork, and software development skills.

TECHNICAL SKILLS

Full Stack Development: React.js, Next.js, Angular, HTML5, CSS3, JavaScript, TypeScript, Node.js, Express.js, FastAPI, Spring Boot, REST APIs

Programming Languages: Java, Python, C, JavaScript, TypeScript, Verilog HDL

Databases: PostgreSQL, MongoDB, MySQL, SQL

AI & Machine Learning: Ollama, LLM Integration, Prompt Engineering, AI Application Development

Networking: TCP/IP, OSI Model, IPv4/IPv6, Subnetting, VLANs, DNS, VPN, Routing, Switching, Network Security, ACL

VLSI & Embedded Systems: RTL Design, Verilog HDL, AMBA Protocols, AHB, APB, AHB-to-APB Bridge, Digital Electronics, Embedded Systems

Semiconductor & EDA: Synopsys Sentaurus TCAD, Semiconductor Device Simulation, TFET Analysis

Tools & Platforms: Git, GitHub, VS Code, Postman, Cisco Packet Tracer, Wireshark, Xilinx Vivado

Cloud & Deployment: Vercel, Netlify, Render

EXPERIENCE

SmartInternz

May 2025 – June 2025

Full Stack Developer (MERN Stack) Intern

Remote

- Worked with MongoDB, Express.js, React.js, and Node.js to develop full-stack web applications.
- Gained hands-on experience in frontend development, REST APIs, database integration, and modern JavaScript workflows.

Maven Silicon

June 2025 – July 2025

Frontend VLSI Design Intern

Remote

- Worked on RTL design and digital hardware development using Verilog HDL and AMBA protocols.
- Designed and verified an AHB-to-APB bridge through RTL implementation, simulation, and waveform analysis.

SkillDzire

December 2025 – March 2026

Web Development and Cloud Integration Intern

Remote

- Worked on web application development using modern frontend and backend technologies.
- Gained exposure to cloud integration, application deployment, and software development workflows.

PROJECTS

PharmaGuard AI – AI-Powered Complaint Management System

GitHub

- **Tech Stack:** React.js, FastAPI, PostgreSQL, SQLAlchemy, Ollama
- Developed a full-stack pharmaceutical complaint management platform with authentication, complaint CRUD, document uploads, REST APIs, and AI-powered complaint analysis.

ELYA AI – Intelligent AI Assistant

GitHub

- **Tech Stack:** Angular, Node.js, Express.js, MongoDB, Ollama
- Built an AI-powered conversational assistant with REST APIs, document understanding, chat history management, and Ollama-based response generation.

Helium Personalization Engine

GitHub

- **Tech Stack:** Next.js, TypeScript, Tailwind CSS, Ollama
- Developed an AI-powered e-commerce personalization application for shopper classification, personalized experiences, and product recommendation workflows.

Product Browser Backend

GitHub

- **Tech Stack:** Python, FastAPI, PostgreSQL, REST APIs
- Designed scalable REST APIs supporting search, filtering, sorting, and pagination across a 200,000+ product dataset with PostgreSQL integration.

ResolveNow – Online Complaint Management System

GitHub

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Socket.IO
- Developed a MERN-based complaint management platform with authentication, role-based access control, complaint tracking, REST APIs, and real-time communication.

ACADEMIC PROJECT

Comparative Analysis of Electrical Characteristics of Si and Ge/Si Double-Gate Tunnel FETs

- **Project Description:** Final-year five-member group project comparing Silicon and Ge/Si Double-Gate Tunnel FETs using Synopsys Sentaurus TCAD for low-power semiconductor applications.
- **Role & Responsibilities:** Acted as a core team member and supported the project lead in device design, TCAD simulation, result analysis, technical documentation, and final presentation.
- **Analysis:** Evaluated transfer characteristics, threshold voltage, subthreshold swing, ON/OFF current ratio, transconductance, and gate capacitance to compare the electrical performance of both device structures.
- **Tools:** Synopsys Sentaurus TCAD, Microsoft Word, Microsoft PowerPoint

EDUCATION

Sasi Institute of Technology and Engineering

2022 – 2026

Bachelor of Technology in Electronics and Communication Engineering

Tadepalligudem, Andhra Pradesh

- CGPA: 7.89 / 10

CERTIFICATIONS & TRAINING

- **SkillDzire** – Full Stack Java Development
- **SkillDzire / APSICHE** – Web Development & Cloud Integration
- **IBM SkillsBuild** – Generative AI in Action
- **AWS Academy / EduSkills** – Cloud Computing Virtual Internship
- **Cisco** – Networking Basics
- **SkillUp by Simplilearn** – Java Programming for Beginners
- **EduSkills / AICTE** – Microchip Embedded System Developer Virtual Internship
- **SkillDzire / AICTE** – VLSI Virtual Internship

SOFT SKILLS

- Problem Solving & Analytical Thinking
- Communication & Team Collaboration
- Leadership, Adaptability & Time Management